

“Calculus 3”

Multi-Variable Calculus

Instructor: Álvaro Lozano-Robledo

Day 17

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Any Reminders? Any Questions?

- Exam on Friday (discussion) covers 15.1-15.3 + 15.6-15.9
- See email message with resources
 - SP26 Practice Exam/Problems!
- Review during Thursday's class
- Office hours Thursday 3:30-4:30 at MONT 233
- Calc Night Thursday 6:30-8:30 at MONT 104

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3

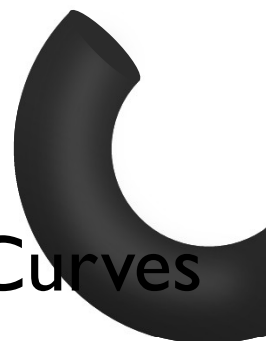
“Calculus 3”



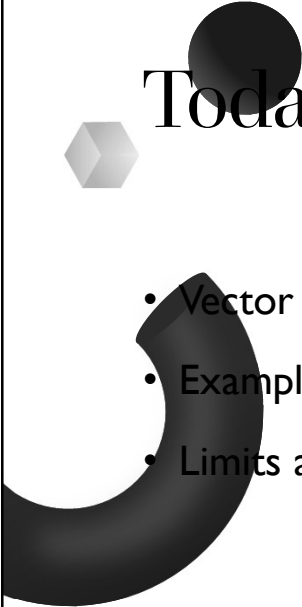
Multi-Variable Calculus

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Vector Functions and Space Curves



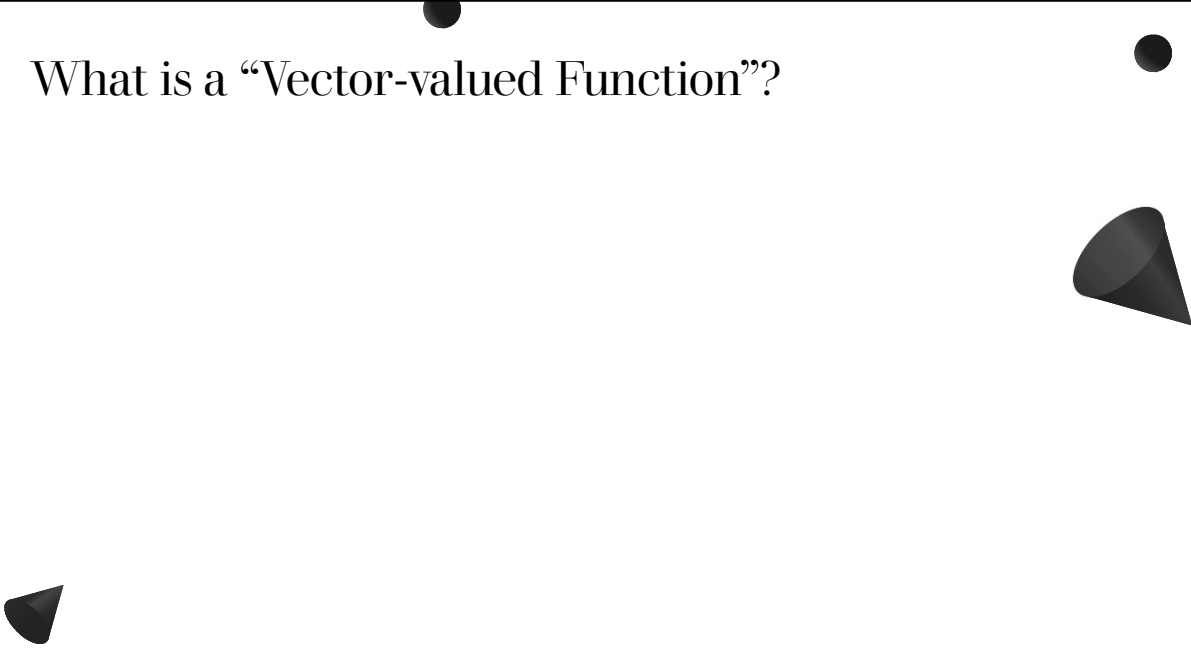
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Today – Vector Functions and Space Curves

- Vector valued functions
- Examples
- Limits and Continuity

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What is a “Vector-valued Function”?

6

Example: Describe the image of the function given by
 $r(t) = (1 + t, 2 + 5t, -1 + 6t)$



7

[Extra Space]

Example: Describe the image of the function given by
 $r(t) = (1 + t, 2 + 5t, -1 + 6t)$



8

Example: Find a vector equation and parametric equation for the line segment that joins $P = (1, -3, 2)$ and $Q = (2, -1, 3)$.

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[Extra Space]

Example: Find a vector equation and parametric equation for the line segment that joins $P = (1, -3, 2)$ and $Q = (2, -1, 3)$.

10

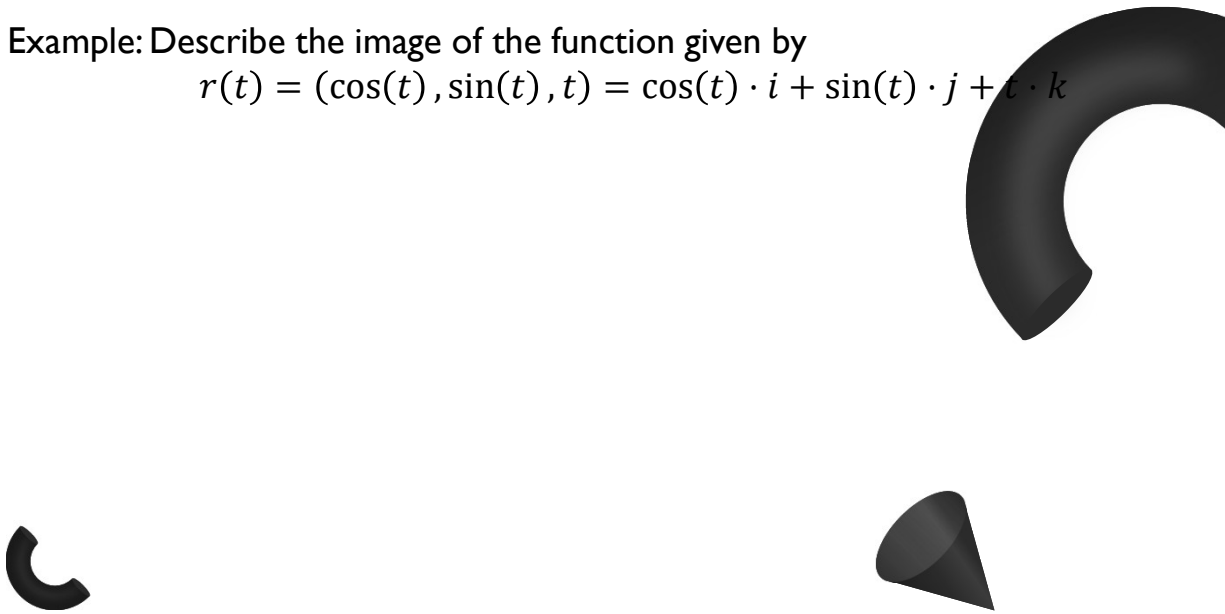
Example: Describe the image of the function given by

$$r(t) = (\cos(t), \sin(t), t) = \cos(t) \cdot i + \sin(t) \cdot j + t \cdot k$$

11

[Extra Space]

Example: Describe the image of the function given by

$$r(t) = (\cos(t), \sin(t), t) = \cos(t) \cdot i + \sin(t) \cdot j + t \cdot k$$


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Example: Find a vector equation that represents the curve at the intersection of $x^2 + y^2 = 1$ and the plane $y + z = 2$.

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Example: Find a vector equation that represents the curve at the intersection of $x^2 + y^2 = 1$ and the plane $y + z = 2$.

[Extra Space]

14

Example: Find a vector equation that represents the curve at the intersection of $4y = x^2 + z^2$ and the plane $y = x$.

15

[Extra Space]

Example: Find a vector equation that represents the curve at the intersection of $4y = x^2 + z^2$ and the plane $y = x$.

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Limits and Continuity of Vector-valued Functions

1 If $\mathbf{r}(t) = \langle f(t), g(t), h(t) \rangle$, then

$$\lim_{t \rightarrow a} \mathbf{r}(t) = \left\langle \lim_{t \rightarrow a} f(t), \lim_{t \rightarrow a} g(t), \lim_{t \rightarrow a} h(t) \right\rangle$$

provided the limits of the component functions exist.

A vector function $\mathbf{r}$ is **continuous at a** if

$$\lim_{t \rightarrow a} \mathbf{r}(t) = \mathbf{r}(a)$$

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Example: Find the limit as t goes to 0:

$$\lim_{t \rightarrow 0} \left(1 + t^3, te^{-t}, \frac{\sin(t)}{t} \right)$$

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Questions?

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Derivatives and Integrals of Vector Functions

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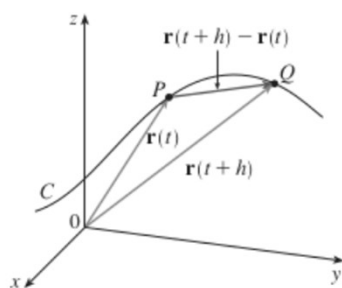
Today – Derivatives and Integrals of Vector Functions

- Derivatives of vector valued functions
- Differentiation rules
- Integrals

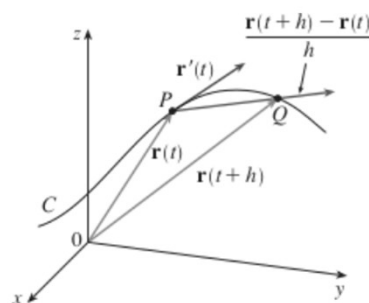
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What is a derivative of a “Vector-valued Function”?

$$\frac{d\mathbf{r}}{dt} = \mathbf{r}'(t) = \lim_{h \rightarrow 0} \frac{\mathbf{r}(t+h) - \mathbf{r}(t)}{h}$$



(a) The secant vector $\overrightarrow{PQ}$



(b) The tangent vector $\mathbf{r}'(t)$

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What is a derivative of a “Vector-valued Function”?

$$\frac{d\mathbf{r}}{dt} = \mathbf{r}'(t) = \lim_{h \rightarrow 0} \frac{\mathbf{r}(t+h) - \mathbf{r}(t)}{h}$$

If $\mathbf{r}(t) = \langle f(t), g(t), h(t) \rangle = f(t) \mathbf{i} + g(t) \mathbf{j} + h(t) \mathbf{k}$, where f , g , and h are differentiable functions, then

$$\mathbf{r}'(t) = \langle f'(t), g'(t), h'(t) \rangle = f'(t) \mathbf{i} + g'(t) \mathbf{j} + h'(t) \mathbf{k}$$

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What is a derivative of a “Vector-valued Function”?

$$\frac{d\mathbf{r}}{dt} = \mathbf{r}'(t) = \lim_{h \rightarrow 0} \frac{\mathbf{r}(t+h) - \mathbf{r}(t)}{h}$$

A unit vector that has the same direction as the tangent vector is called the **unit tangent vector** $\mathbf{T}$ and is defined by

$$\mathbf{T}(t) = \frac{\mathbf{r}'(t)}{|\mathbf{r}'(t)|}$$

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Example: Find the derivative of the function given by
 $\mathbf{r}(t) = (1 + t, 2 + 5t, -1 + 6t)$
 and find the unit tangent vector at $t = 2$.

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[Extra Space]

Example: Find the derivative of the function given by

$$r(t) = (1 + t, 2 + 5t, -1 + 6t)$$

and find the unit tangent vector at $t = 2$.

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Example: Find the derivative of the function given by

$$r(t) = (\cos(t), \sin(t), t) = \cos(t) \cdot i + \sin(t) \cdot j + t \cdot k$$

and find the unit tangent vector at $t = \frac{\pi}{2}$.

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[Extra Space]

Example: Find the derivative of the function given by

$$\mathbf{r}(t) = (\cos(t), \sin(t), t) = \cos(t) \cdot \mathbf{i} + \sin(t) \cdot \mathbf{j} + t \cdot \mathbf{k}$$

and find the unit tangent vector at $t = \frac{\pi}{2}$.

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Differentiation Rules for Vector-valued Functions

Suppose $\mathbf{u}$ and $\mathbf{v}$ are differentiable vector functions, c is a scalar, and f is a real-valued function.

Then

1. $\frac{d}{dt}[\mathbf{u}(t) + \mathbf{v}(t)] = \mathbf{u}'(t) + \mathbf{v}'(t)$
2. $\frac{d}{dt}[c\mathbf{u}(t)] = c\mathbf{u}'(t)$
3. $\frac{d}{dt}[f(t)\mathbf{u}(t)] = f'(t)\mathbf{u}(t) + f(t)\mathbf{u}'(t)$
4. $\frac{d}{dt}[\mathbf{u}(t) \cdot \mathbf{v}(t)] = \mathbf{u}'(t) \cdot \mathbf{v}(t) + \mathbf{u}(t) \cdot \mathbf{v}'(t)$
5. $\frac{d}{dt}[\mathbf{u}(t) \times \mathbf{v}(t)] = \mathbf{u}'(t) \times \mathbf{v}(t) + \mathbf{u}(t) \times \mathbf{v}'(t)$
6. $\frac{d}{dt}[\mathbf{u}(f(t))] = f'(t)\mathbf{u}'(f(t))$ (Chain Rule)

If $|\mathbf{r}(t)| = c$ (a constant), then $\mathbf{r}'(t)$ is orthogonal to $\mathbf{r}(t)$ for all t .

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Integrals of “Vector-valued Functions”

$$\begin{aligned}\int_a^b \mathbf{r}(t) dt &= \lim_{n \rightarrow \infty} \sum_{i=1}^n \mathbf{r}(t_i^*) \Delta t \\ &= \lim_{n \rightarrow \infty} \left[\left(\sum_{i=1}^n f(t_i^*) \Delta t \right) \mathbf{i} + \left(\sum_{i=1}^n g(t_i^*) \Delta t \right) \mathbf{j} + \left(\sum_{i=1}^n h(t_i^*) \Delta t \right) \mathbf{k} \right]\end{aligned}$$

and so

$$\int_a^b \mathbf{r}(t) dt = \left(\int_a^b f(t) dt \right) \mathbf{i} + \left(\int_a^b g(t) dt \right) \mathbf{j} + \left(\int_a^b h(t) dt \right) \mathbf{k}$$

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Example: Find the integral

$$\int_0^{\pi/2} \mathbf{r}(t) dt$$

of the function given by

$$\mathbf{r}(t) = (\cos(t), \sin(t), t) = \cos(t) \cdot \mathbf{i} + \sin(t) \cdot \mathbf{j} + t \cdot \mathbf{k}$$

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[Extra Space]

Example: Find the integral

$$\int_0^{\pi/2} r(t) dt$$

of the function given by

$$r(t) = (\cos(t), \sin(t), t) = \cos(t) \cdot i + \sin(t) \cdot j + t \cdot k$$

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Questions?

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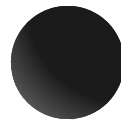


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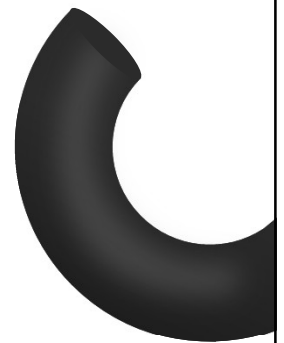


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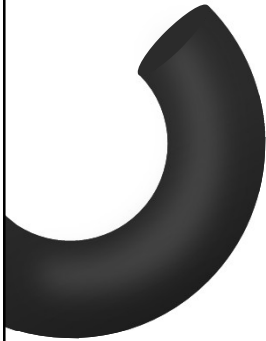
Arc Length



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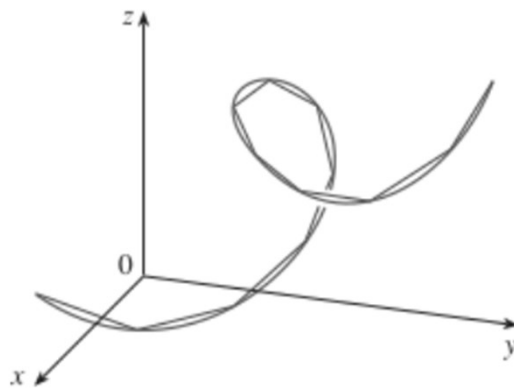
Today – Arc Length



- Length of a segment of a curve...
 - In the plane
 - In 3d-space

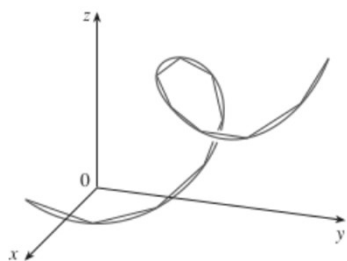
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How do we compute the length of a curve?



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How do we compute the length of a curve?



$$L = \int_a^b \sqrt{[f'(t)]^2 + [g'(t)]^2} dt = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$$

$$\begin{aligned} L &= \int_a^b \sqrt{[f'(t)]^2 + [g'(t)]^2 + [h'(t)]^2} dt \\ &= \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2} dt \end{aligned}$$

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How do we compute the length of a curve?

$$L = \int_a^b |\mathbf{r}'(t)| dt$$

because, for plane curves $\mathbf{r}(t) = f(t) \mathbf{i} + g(t) \mathbf{j}$,

$$|\mathbf{r}'(t)| = |f'(t) \mathbf{i} + g'(t) \mathbf{j}| = \sqrt{[f'(t)]^2 + [g'(t)]^2}$$

and for space curves $\mathbf{r}(t) = f(t) \mathbf{i} + g(t) \mathbf{j} + h(t) \mathbf{k}$,

$$|\mathbf{r}'(t)| = |f'(t) \mathbf{i} + g'(t) \mathbf{j} + h'(t) \mathbf{k}| = \sqrt{[f'(t)]^2 + [g'(t)]^2 + [h'(t)]^2}$$

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Example: Find the length of the curve given by

$$r(t) = (\cos(t), \sin(t), t) = \cos(t) \cdot i + \sin(t) \cdot j + t \cdot k$$

from the point $P = (1,0,0)$ to the point $Q = (1,0,2\pi)$.

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[Extra Space]

Example: Find the length of the curve given by

$$r(t) = (\cos(t), \sin(t), t) = \cos(t) \cdot i + \sin(t) \cdot j + t \cdot k$$

from the point $P = (1,0,0)$ to the point $Q = (1,0,2\pi)$.

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Example: Find the length of the curve given by

$$r(t) = (2t, t^2, \frac{t^3}{3})$$

from $t = 0$ to $t = 1$.



43

[Extra Space]

Example: Find the length of the curve given by

$$r(t) = (2t, t^2, \frac{t^3}{3})$$

from $t = 0$ to $t = 1$.



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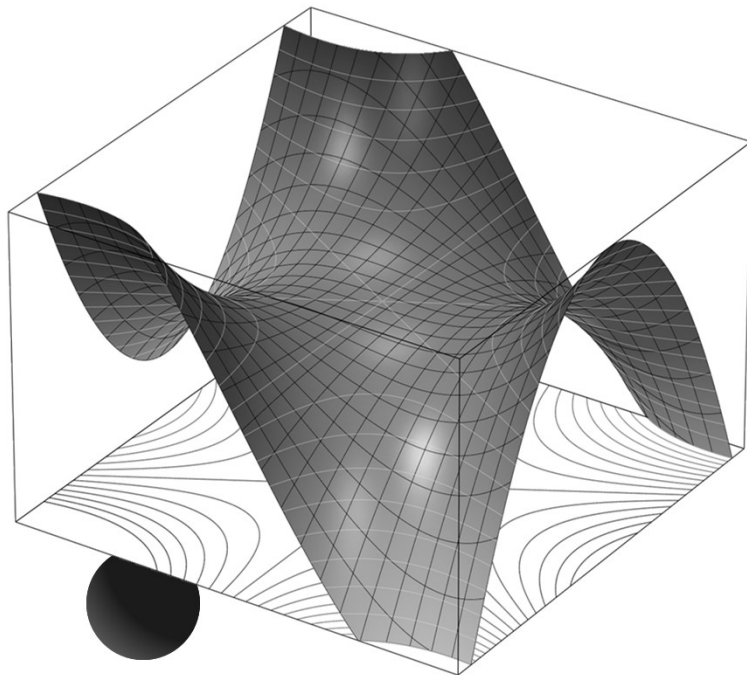
Questions?



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Thank you

Until next time.



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